

Medial Axis Extraction and Tracking for Metallic Tools in Video Sequences

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Abstract

This project IMPLEMENTS an advanced computer vision pipeline to detect, track, and extract the medial axis of a metallic tool across multiple video sequences. By combining Background Subtraction (MOG2), frame differencing, and HSV color filtering, the system isolates moving metallic objects from complex backgrounds. The core contribution is a robust Medial Axis extraction method using a custom Hough Line Transform and angular grouping. In the context of Aerial Robotics Kharagpur (ARK), this can be scaled for autonomous robotic grasping, where identifying the centerline (medial axis) of an object is critical for calculating optimal grip points and orientation.

I. INTRODUCTION

The challenge addressed in this project is the consistent tracking of a tool and the mathematical determination of its orientation (centerline) in a video feed. Simple color-based tracking fails due to changing lighting and metallic reflections. Our approach utilizes a "triple-mask" method (static background, frame difference, and MOG2) to ensure the foreground (FG) remains stable. We then apply a custom Hough Transform to the tool's edges to find the medial axis. We attempted simple centroid tracking, but it failed to provide orientation data, leading to the development of the Hough-based medial axis algorithm.

II. PROBLEM STATEMENT

The task involves processing three video files (1.mp4, 2.mp4, 3.mp4) to identify a specific tool and draw its medial axis in real-time.

- **Detection:** Separating a moving metallic object from a potentially moving or noisy background.
- **Tracking:** Maintaining a "region of interest" (ROI) even when the tool is partially obscured or noise increases.
- **Geometric Analysis:** Extracting the centerline of the tool regardless of its rotation or scale within the frame.

III. RELATED WORK

Traditional object tracking often uses Optical Flow (Lucas-Kanade) or simple Background Subtraction. However, these do not provide the geometric skeleton of the object. Medial Axis Transform (MAT) is usually done via distance transforms, but in noisy video, these are highly sensitive to "jagged" edges. Our approach uses Hough Line grouping [1] to find the

dominant linear axis, which is more robust to boundary noise in industrial/robotic environments.

IV. INITIAL ATTEMPTS

Our initial attempt used a simple Hough Circle transform to find the tool's joints, but metallic glare created too many false positives. We also tried a basic `cv2.distanceTransform` for the medial axis, but the resulting "skeleton" was too noisy and branched off into many segments, making it impossible to define a single orientation vector for robotic path planning.

V. FINAL APPROACH

The final algorithm follows a multi-stage pipeline:

1. Multi-Stage Foreground Detection

The system generates a mask by OR-ing three distinct sources:

- **Static BG:** Absolute difference between the frame and a pre-computed median background.
- **Frame Diff:** Difference between the current and previous frame to catch fast motion.
- **MOG2:** A Gaussian Mixture-based subtractor to handle subtle background variations.
- **Saturation Filter:** Since the tool is metallic (low saturation), we use an HSV filter to remove high-saturation noise.

2. Tool Isolation (best_tool function)

We filter contours based on:

- **Area:** $\text{Area} > 300$ pixels.
- **Aspect Ratio:** $\text{Aspect Ratio} > 1.5$ (ensures the object is elongated, like a tool).
- **Scoring:** A custom score $S = \text{aspect} \times \sqrt{\text{area}}$ is used to pick the most likely candidate.

3. Medial Axis Extraction

Within the tool's bounding box:

- **Edge Detection:** Sobel operators find the tool's boundaries.
- **Custom Hough:** `hough_lines` calculates the accumulation matrix for (ρ, θ) .

- **Grouping:** Lines are grouped by angle ($\Delta\theta < 15^\circ$). The two most dominant lines representing the tool's sides are averaged to find the medial axis.

VI. RESULTS AND OBSERVATION

The algorithm was tested on three videos with the following observations:

Video	Performance	Observation
1.mp4	Excellent	High contrast allowed for perfect medial axis extraction.
2.mp4	Good	Slight jitter when the tool moved too close to background shadows.
3.mp4	Stable	The "boost" near last known position kept the track even during low-motion segments.

Drawbacks: The Hough accumulation is computationally expensive ($O(N \cdot \theta)$). However, by limiting the search to the bounding box (bbox), we maintain near real-time performance.

VII. FUTURE WORK

Currently, the θ resolution is fixed at 1° . Future work could implement a pyramid-style Hough transform (coarse-to-fine) to increase precision without increasing computation. Additionally, integrating a Kalman Filter would allow the system to predict the medial axis position during complete occlusions.

CONCLUSION

We successfully developed a tool-tracking system that extracts the medial axis from video frames. By using a "Triple Mask" foreground detection and angular-grouped Hough lines, the system overcomes the common pitfalls of metallic reflections and background noise. For ARK, this provides a reliable way to determine object orientation, enabling more precise autonomous interaction with tools and components.